

🌀 M003 Extension — The Loop Pattern (Beginner)

Topic: One loop, three counters (x, y, z) · **Course:** 3D Coordinates · **Level:** Extension (Beginner, after Week 6) · **Time:** about 30 minutes

Until now you placed blocks by choosing **every** coordinate **by hand**. This time **one loop** builds all **7** obsidian walls for you. The loop keeps three counters — **x, y, z** — and changes each one a little every pass. That is an **algorithm**: a short list of steps a computer repeats. Load the world, type **SPIRAL** in the chat, and watch 7 walls grow into a pattern.

```
x = 14
y = 0
z = 2
for step in range(7):
    fill OBSIDIAN from (14, 0, z) to (x, y, z)
    x = x - 2
    y = y + 1
    z = z + 2
```

 **Big idea:** the loop does the same three changes every pass — x goes **down 2**, y goes **up 1**, z goes **up 2**. So each new wall is a little **wider, taller, and deeper** than the last.

 Stand on your **home spot** (red block) every run.

 Every wall is **obsidian**.

1 · Predict 🌟

Read the code. Circle one.

for step in range(7) — how many walls does the loop build? Circle one: 3 · 7 · 10

Each pass runs **y = y + 1**. Does that make the next wall...? Circle one: taller · wider · deeper

Each pass runs **z = z + 2**. Does that make the next wall...? Circle one: taller · wider · deeper

The loop starts with z = 2. After one pass, **z = z + 2** runs. What is z now? Circle one: 3 · 4 · 6

2 · Spot the Bug 🐛

The `z = z + 2` line was deleted. Now `z` stays **2** every pass, so all 7 walls land in the **same spot**, on top of each other — you see one wall, not a pattern.

```
for step in range(7):  
    fill OBSIDIAN from (14, 0, z) to (x, y, z)  
    x = x - 2  
    y = y + 1
```

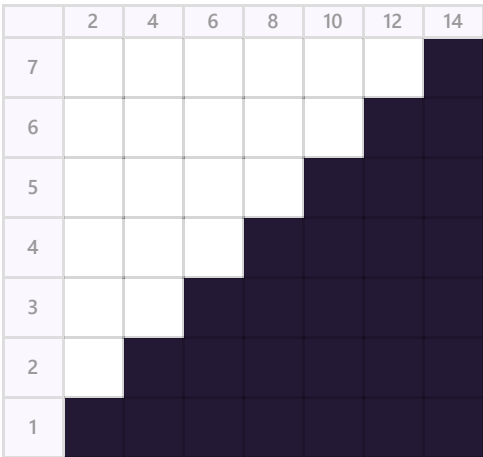
What is missing? Circle one: the `z` step · the loop

Should each wall sit at a new `z` or the same `z`? Circle one: a new `z` · the same `z`

3 · Show Your Work 📷 🎥

Load the world. Stand on your home spot. Type **SPIRAL** in the chat and watch the loop build all 7 walls. Walk around the pattern.

🧱 The finished pattern — side view (y ↑ up, z → deeper)



Reading left to right, the steps get taller: 1, 2, 3, 4, 5, 6, 7.

How many steps are there? Circle one: 5 · 7 · 9

Which step is the tallest? Circle one: the first (front) · the last (back)

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · **Your code.** Scroll through it. Say what each part does.

2 · **Your build.** Point the camera at the pattern. Name the smallest and biggest step.

Say these out loud, filling in the blanks:

Today I built ____.

I built it using this code: ____.

In this code I used ____.

The hardest part was ____.

That part was hard because ____.

The most fun part was ____.

Something new I learned was ____.

Submit ✓

Send this worksheet + your **one** video to teacher on KakaoTalk.